



Independent Watchdog (IWDG) Overview

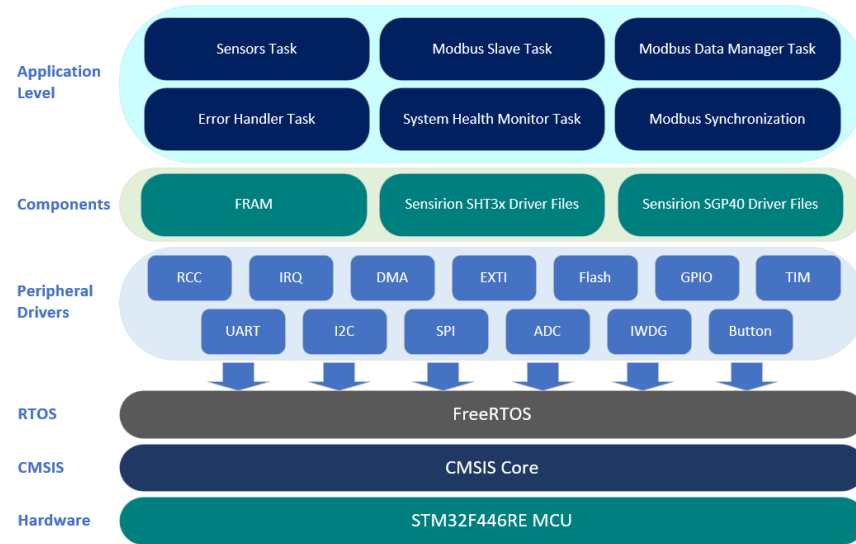
Overview & Implementation Steps

Independent Watchdog (IWDG) Overview

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Independent Watchdog (IWDG) Overview

- **Definition:** The **Independent Watchdog (IWDG)** is a hardware timer that resets the system if not periodically reset.
- **Purpose & Functionality:**
 - **Safety Mechanism:** Provides a fail-safe mechanism by resetting the MCU if the application becomes unresponsive.
 - **System Reliability:** Ensures system stability by preventing indefinite hangs.
- **Project Significance:**
 - **System Health Monitor:** Resets the IWDG only if the system is `healthy`, meaning all critical tasks are running and the temperature is within an acceptable range.



IWDG Use in the Project

Overview

- **Definition:** The Independent Watchdog (IWDG) is a timer that resets the system if it is not periodically reset, ensuring that the system does not hang.
- **Functionality & Reliability:** The IWDG operates independently of the main system clock and remains active even if the main clock fails.

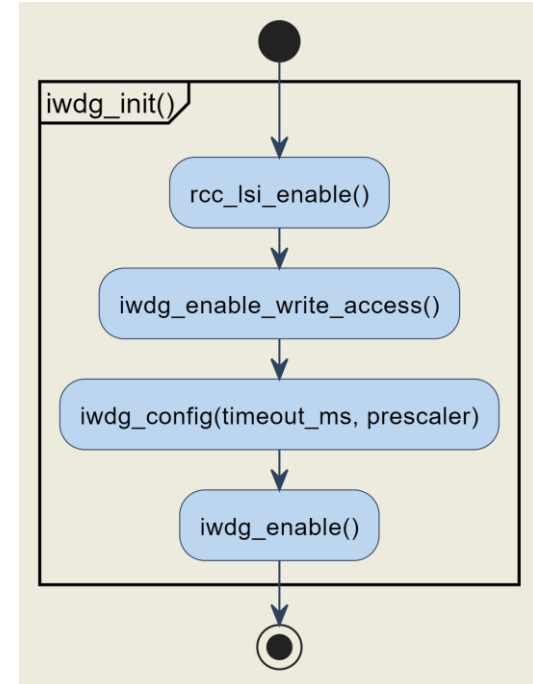
Key Points

- **Flexibility:** Suitable for applications requiring a watchdog to run independently outside the main application but with lower timing accuracy constraints.
- **Application Fit:** The IWDG is ideal for our project since it provides robust system monitoring and recovery, critical for maintaining reliability of the application.
- **Simplicity:** Simpler to configure and use compared to the WWDG, which requires more complex timing window settings.

IWDG Initialization Function

Initialization Steps

- **Enable LSI:** Activates the Low-Speed Internal clock, providing a clock source for the IWDG.
- **Enable Write Access:** Allows modifications to the IWDG registers by enabling write access.
- **IWDG Configuration:** Configures the IWDG prescaler and reload values to set the desired timeout period.
- **IWDG Enable:** Initiates the IWDG by writing to its key register, starting the watchdog timer.



IWDG Configuration Function

Purpose

- **Timeout Configuration:** Allows for specifying the timeout and timeout range, specified by the prescaler:

```
static void iwdg_config(uint16_t timeout_ms, iwdg_prescaler_e prescaler);
```

Key Points

- **Prescaler Selection:** Converts enumeration value to actual prescaler divisor and sets the maximum allowable timeout for each prescaler.
- **Timeout Validation:** Ensures desired timeout does not exceed the maximum allowed for the chosen prescaler.
- **Reload Value Calculation:** Calculates the raw reload value based on desired timeout, LSI frequency, and chosen prescaler.
- **Setting Values:** Sets the prescaler and reload value in the IWDG registers.

Practical Implementation Steps (Header File)

Key Points

- **Key Definitions:** Define essential keys for enabling, writing access, and resetting the IWDG.
- **Enumeration of IWDG Prescalers:** Define possible prescaler values for the IWDG.
- **Static Inline Functions:** Provide inline functions for enabling, resetting, and allowing write access to the IWDG.
- **Public Initialization Function:** Prototype for the IWDG initialization function to configure and start the watchdog.

Importance

- **Interface Provision:** Provides a clear interface for the source file, ensuring modularity and ease of use.

Practical Implementation Steps (Source File)

Key Points

- **Define Public IWDG Initialization Function**
 - Call functions to enable LSI, enable write access, configure the IWDG, and start the IWDG.
- **Define the IWDG Configuration Function**
 - Convert the desired timeout and prescaler value to register values.
 - Validate the timeout value to ensure it does not exceed the maximum for the chosen prescaler.
 - Calculate and set the prescaler and reload values for the IWDG timer.

Importance

- Provides a single entry-point to initialize the IWDG.

The background features a complex network of thin, light-colored lines and small circles, forming various triangular shapes. Some of these shapes are filled with a very light yellow or green color. The overall effect is a subtle, geometric pattern that resembles a wireframe or a network diagram.

Next: IWDG Header File
